A self-refreshing attractor account of long-term memory persistence under molecular turnover

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Document status: public-facing research draft. This document defines testable hypotheses and model-comparison logic. It is not clinical guidance, not a diagnosis, not a claim of proof of consciousness, and not a metaphysical or theological claim.

Core question

If a memory can remain available for decades while the physical molecular components of neurons are continuously replaced, what exactly persists? This theory addresses the Ship-of-Theseus constraint for any structural-substrate account of memory.

Central thesis

A long-term memory should not be modeled as the survival of the same molecules. It should be modeled as a self-refreshing attractor distributed across structural state, scaffold stabilization, synaptic/circuit activity, transcriptional or epigenetic bias, and replay/reactivation.

$$M(t) = [\ P(t), B(t), W(t), Z(t), R(t)\ ]$$

Variable	Public meaning	Permitted claim
P(t)	candidate structural or cytoskeletal trace strength	may represent a structural contribution to memory if independently measured
B(t)	stabilization/scaffold state	may reduce effective turnover or drift
W(t)	synaptic and circuit weight pattern	standard neural plasticity layer
Z(t)	transcriptional/epigenetic regulatory bias	workshop settings that bias future rebuilding, not a memory video file
R(t)	replay, recall, cue reactivation, or activity-dependent refresh	possible mechanism that re-enters the attractor and maintains the trace

Refresh ratio

$$R\_memory = K\_refresh / lambda\_eff$$

The memory trace can persist only when the refresh gain exceeds the effective turnover and drift loss.

$$R\_memory > 1 \implies \text{trace persistence is possible}$$

$$R\_memory < 1 \implies \text{trace decay is expected}$$

Fidelity condition

Persistence alone is insufficient. A refreshed trace must also remain similar enough to the original functional pattern.

$$delta^* = (\text{copy error} + \text{noise}) / K\_correct < \text{identity threshold}$$

Why this matters for consciousness research

Many structural or microtubule-adjacent theories of memory face an immediate biological objection: proteins turn over. This theory converts that objection into a testable constraint. Any structural-memory model must explain pattern maintenance despite substrate replacement.

Non-claims

- This theory does not claim that memory is proven to be microtubular or quantum.
- It does not claim that DNA stores exact memories.
- It does not claim that scalp EEG can verify molecular refresh.
- It does not replace synaptic plasticity; it embeds structural hypotheses inside a multi-layer memory-maintenance model.

Falsification logic

- If structural traces cannot be measured, the structural component P is unsupported.
- If candidate traces persist while all proposed refresh channels are disrupted, the model is incomplete or wrong.
- If transcriptional/epigenetic bias does not predict trace-specific replacement fidelity, the Z component is weakened.
- If the model cannot outperform simple synaptic-only or replay-only models, it should be treated as unnecessary.